Kunal Prajapati

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SUMMARY

I am a skilled medical physicist with expertise in radiation therapy, proton therapy, and dosimetry. With advanced qualifications in radiological physics and experience in treatment planning, quality assurance, and system troubleshooting, I am committed to advancing oncological treatments through innovation and research-driven solutions.

EDUCATION

Homi Bhabha National Institute, BARC — Aug 2020
Post MSc. Diploma in Radiological Physics — 85.17%

Indian Institute of Technology, Delhi — May 2019
Master of Science (MS) in Physics — CGPA 7.725

Panjab University — May 2016
Bachelor of Science (BS) — 78.55%

CERTIFICATIONS

- Radiation safety officer (RSO), level III, *Atomic Energy Regulatory Board, India* — Dec 2022
- CS50's Introduction to Programming with Python, *Harvard University* — Dec 2024
- Python for Data Science, *National Program on Technology Enhanced Learning (NPTEL)* — Aug 2022
- Accelerator Physics, *National Program on Technology Enhanced Learning (NPTEL)* — Oct 2022
- Medical Image Analysis, *National Program on Technology Enhanced Learning (NPTEL)* — Oct 2022
- Good clinical practice, *NIDA Clinical Trials Network* — Jan 2023

SKILLS

- **Technical:**
  Machine learning | Python/Scikit-learn | Data Visualization | SPSS
  MATLAB (beginner) | Origin | MS Office | 3D Printing
- **Radiotherapy Planning:**
  Varian Eclipse TPS | Elekta Oncentra TPS | TomoTherapy Hi-ART TPS
- **Languages:**
  English (Fluent) | Hindi (Native) | Punjabi (Fluent)

WORK EXPERIENCE

IBA Proton Therapy, TMC Mumbai — Navi Mumbai
Beam Physics Service Engineer cum Radiation Safety Officer — May 2023 – Present

- Maintained and deployed the Proton Therapy System (PTS) for QA, development, testing, troubleshooting, and research support.
- Calibrated beam-related equipment and supported clinical operations, ensuring safe and accurate beam delivery.
- Contributed to radiation safety compliance within clinical and experimental settings.

Advanced Centre for Treatment, Research, and Education in Cancer (ACTREC) Navi Mumbai
Medical Physicist Feb 2022 – May 2023

- Conducted treatment planning and QA for advanced radiotherapy techniques like VMAT, TBI, TSET, TMLI.
- Contributed to clinical research in Medical Physics and Radiation Oncology.
- Performed HDR Brachytherapy planning and QA.

Tata Memorial Hospital, Mumbai Mumbai
Resident Medical Physicist Jan 2021 – Dec 2021

- Conducted treatment planning and QA for different radiotherapy techniques.
- Carried out HDR brachytherapy planning and QA.

HANDS-ON EXPERIENCE WITH RADIOTHERAPY SYSTEMS

- **Proton Therapy System**
  IBA Proteus PLUS
- **LINACs**
  Varian Truebeam Varian Novalis Tx Varian Trilogy Accuray Tomotherapy
- **Brachytherapy Units**
  Elekta microSelectron V3 afterloading system Elekta Flexitron afterloading system

PUBLICATIONS

- **Sigmoid dose accumulation and reporting for multifractionated brachytherapy for cervical cancer**
  Bindal, M. A., Mittal, P., Shinghal, A., Scaria, L., **Prajapati, K**., et al. (2021)
- **Patterns of relapse after adjuvant (chemo)radiation for cervical cancer in a phase III clinical trial (PARCER): an evaluation of updated NRG Oncology /RTOG target delineation guidelines**
  Mittal, P., Chopra, S., Charnalia, M., Dora, T., Engineer, R., **Prajapati, K.**, et al. (2022)

ABSTRACTS

- **Dosimetric comparison of 1.0 mm vs 2.5mm source step size in treatment plan quality of Breast brachytherapy**
  Prajapati, K., Scaria, L., et al. (2022)
- **Validation of sigmoid points for sigmoid dose accumulation and reporting for multifractionated brachytherapy for cervical cancer**
  Kamalnath, J., A. K., **Prajapati, K.**, et al. (2024)
- **Sigmoid Points predict Late Toxicity Assessment for Multifractionated Brachytherapy for Cervical Cancer**
  Mittal P., Kamalnath, J., A. K., **Prajapati, K.**, et al. (2024)

PROFESSIONAL AFFILIATIONS

Association of Medical Physicists in India (AMPI), *Medical Physicist* Life-time member